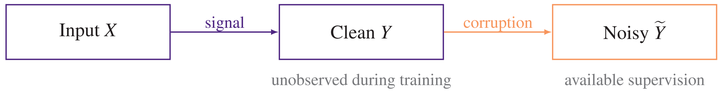
Dual-Granularity Learning for Regression with Continuous Noisy Labels

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Problem Setup

- **Regression with Continuous Noisy Labels:** learn a continuous-valued predictor from labels corrupted by heterogeneous, potentially instance-dependent noise
- **Data and Model:** only noisy labels are observed:

$$\mathscr{D} = \{(X_i, \widetilde{Y}_i)\}_{i=1}^N, \qquad X_i \in \mathbb{R}^p, \quad \widetilde{Y}_i \in \mathbb{R}$$
$$Y = f(X;\theta^*) + \varepsilon', \qquad \widetilde{Y} = Y + \varsigma(X,Y)\varepsilon''$$



- $f(\cdot;\theta^*)$: the unknown ground-truth regression function
- ε' and ε'': independent zero-mean noise variables with finite variance
- $\varsigma(X,Y)$ allows the noise magnitude to vary across inputs and labels
- **Goal:** recover predictive signals, with the effect of label corruption addressed, so that the model generalizes to unseen inputs

Regression Differs from Classification: Labels are continuous and have complex structures. The proposed GRAIN incorporates these characteristics at different granularities.

Motivation

- **Heterogeneous Label Noise Creates Conflicting Requirements.** A single correction mechanism cannot simultaneously fully address both local numeric deviations and global feature distortion

Challenge 1: Precise Local Correction

Small-to-moderate deviations retain useful numeric information. Correct them through **instance-specific residual learning**. ⟷ Fine-Grained Refinement

Challenge 2: Robust Global Structure

For large deviations, exact labels become unreliable. Preserve label proximity through **order-aware structural regularization**. ⟷ Coarse-Grained Structuring

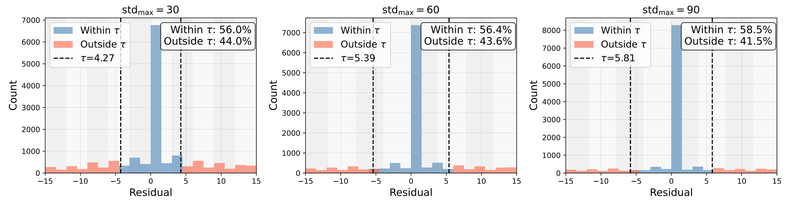


Figure 1: True label-noise distributions at the final training epoch on UTKFace under three noise levels. Dashed lines mark the residual capacity $\pm\tau$.

GRAIN: Theoretical Analysis and Methodology

A Dual-Granularity Framework for Noisy Regression

Explicit residual learning | Order-preserving feature geometry | Histogram-based bridging

1. Fine-Grained Refinement: Loss-Induced Residuals

The piecewise robust loss preserves local sensitivity while controlling the influence of large deviations:

$$\ell_\tau(v) = \begin{cases} \frac{1}{2}v^2, & |v| \le \tau \\ \frac{\tau}{\ell'(\tau)}\{\ell(|v|) - \ell(\tau)\} + \frac{\tau^2}{2}, & |v| > \tau \end{cases}$$

Proposition (Residual Reformulation).

$$\ell_\tau(v) = \inf_{\xi\in\mathbb{R}} \left\{\tfrac{1}{2}\xi^2 + \phi_\tau(v-\xi)\right\}$$

Capacity-Constrained Residual Head:

$$\xi_i = \tau\tanh\big(h_{\text{res}}(Z_i;\theta_{\text{res}})\big) \in (-\tau,\tau) \text{ with } Z_i = \varphi(X_i;\theta_{\text{f}}).$$

Its residual target is $\widetilde{Y}_i - Y_i^*$, where Y_i^* is a robust pseudo label; τ is updated using a median absolute deviation (MAD)-based residual scale.

Theorem. Assume $Y = \langle Z, \theta^*_{\text{reg}}\rangle + \varepsilon'$ for a learned feature map $Z = \varphi(X) \in \mathbb{R}^d$, and let $\widehat{\theta}^\tau_{\text{reg}}$ denote the minimizer of the empirical risk $\mathscr{R}^\tau_{\text{reg}} \triangleq \frac{1}{N}\sum_{i=1}^N \ell_\tau(\langle Z_i, \theta_{\text{reg}}\rangle - \widetilde{Y}_i)$. Under regularity conditions, for any $s > 0$, with probability at least $1 - 5e^{-s}$,

$$\|\widehat{\theta}^\tau_{\text{reg}} - \theta^*_{\text{reg}}\|_\Sigma \le C\sqrt{\frac{d+s}{N}}.$$

2. Coarse-Grained Structuring across Bins

Density-adaptive binning defines bin-wise feature distributions $\widehat{P}_Z^{(k)}$. Nearby bins ($\mathscr{N}_k^+$) are pulled together; farther bins ($\mathscr{N}_k^-$) are pushed apart:

$$\mathscr{R}_{\text{pos}} = \frac{1}{K}\sum_k \sum_{k'\in\mathscr{N}_k^+} w^+_{kk'}\mathsf{d}\left(\widehat{P}_Z^{(k)}, \widehat{P}_Z^{(k')}\right); \quad \mathscr{R}_{\text{neg}} = -\frac{1}{K}\sum_k \sum_{k'\in\mathscr{N}_k^-} w^-_{kk'}\mathsf{d}\left(\widehat{P}_Z^{(k)}, \widehat{P}_Z^{(k')}\right)$$

where $w^+_{kk'} \propto 1/|k-k'|$ and $w^-_{kk'} \propto (k-k')^2$, normalized within their neighborhoods.

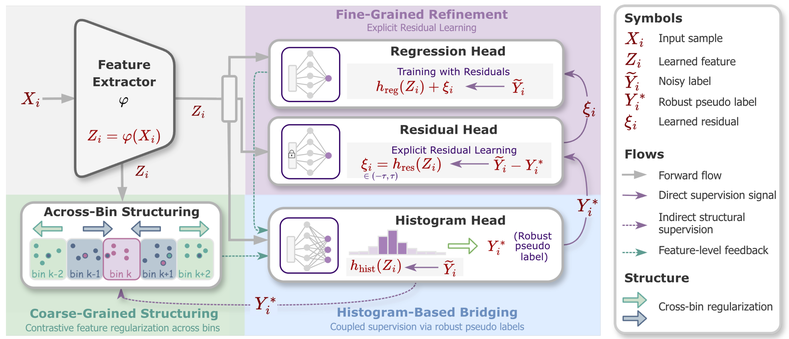


Figure 2: Workflow of GRAIN. Fine-grained refinement (top right), Coarse-grained structuring (bottom left), and Histogram-based bridging (bottom right).

3. Histogram-Based Bridging

The histogram head generates robust pseudo labels:

$$Y_i^{*(t)} = \alpha Y_i^{*(t-1)} + (1-\alpha)\frac{\sum_{k\in T_i^{(t)}} p_i^{(t)}[k]\,\bar{y}_k}{\sum_{k\in T_i^{(t)}} p_i^{(t)}[k]}$$

Here, $\bar{y}_k$ is the bin representative. Starting from $Y_i^{*(0)} = \widetilde{Y}_i$ after warm-up, we use $\alpha = 0.9$. $T_i^{(t)}$ retains the highest-probability bins until their cumulative mass reaches a threshold, say 0.68

- **Global structuring:** Y_i^* stabilizes bin assignments under noisy supervision
- **Local correction:** $\widetilde{Y}_i - Y_i^*$ provides residual targets for the fine-grained module
- **Feedback:** refined, ordered features improve histogram predictions, yielding more reliable pseudo labels in subsequent epochs

Joint objective: $\mathscr{L} = \lambda_{\text{reg}}\mathscr{R}^\tau_{\text{reg}} + \lambda_{\text{ce}}\mathscr{R}_{\text{ce}} + \lambda_{\text{pos}}\mathscr{R}_{\text{pos}} + \lambda_{\text{neg}}\mathscr{R}_{\text{neg}}$.

Experimental Results

Comparison with Baseline Methods

Method	UTKFace (std$_{max}$ = 30)			UTKFace (std$_{max}$ = 60)			UTKFace (std$_{max}$ = 90)		
	MAE ↓	RMSE ↓	R^2 ↑	MAE ↓	RMSE ↓	R^2 ↑	MAE ↓	RMSE ↓	R^2 ↑
MSE (w/o binning)	$8.13_{\pm 0.17}$	$11.20_{\pm 0.12}$	$0.70_{\pm 0.01}$	$9.94_{\pm 0.19}$	$13.07_{\pm 0.22}$	$0.59_{\pm 0.01}$	$10.25_{\pm 0.29}$	$13.43_{\pm 0.31}$	$0.56_{\pm 0.02}$
CE (with binning)	$7.79_{\pm 0.09}$	$11.87_{\pm 0.12}$	$0.66_{\pm 0.01}$	$8.44_{\pm 0.17}$	$13.12_{\pm 0.30}$	$0.58_{\pm 0.02}$	$8.46_{\pm 0.28}$	$13.06_{\pm 0.44}$	$0.59_{\pm 0.03}$
Best (w/o binning)	$7.56_{\pm 0.16}$	$10.40_{\pm 0.24}$	$0.74_{\pm 0.01}$	$7.98_{\pm 0.09}$	$11.95_{\pm 0.32}$	$0.66_{\pm 0.02}$	$8.21_{\pm 0.22}$	$11.97_{\pm 0.33}$	$0.65_{\pm 0.02}$
Best (with binning)	$7.44_{\pm 0.25}$	$10.40_{\pm 0.36}$	$0.74_{\pm 0.02}$	$7.90_{\pm 0.19}$	$11.61_{\pm 0.31}$	$0.67_{\pm 0.02}$	$7.95_{\pm 0.19}$	$11.65_{\pm 0.26}$	$0.67_{\pm 0.01}$
GRAIN	$7.01_{\pm 0.10}$	$10.28_{\pm 0.24}$	$0.74_{\pm 0.01}$	$7.54_{\pm 0.14}$	$11.10_{\pm 0.33}$	$0.70_{\pm 0.02}$	$7.65_{\pm 0.12}$	$11.28_{\pm 0.22}$	$0.70_{\pm 0.01}$
GRAIN-AUG	$6.81_{\pm 0.13}$	$9.98_{\pm 0.28}$	$0.76_{\pm 0.01}$	$7.30_{\pm 0.14}$	$10.74_{\pm 0.28}$	$0.72_{\pm 0.01}$	$7.35_{\pm 0.22}$	$10.75_{\pm 0.36}$	$0.72_{\pm 0.02}$

Table 1: Mean ± standard deviation on UTKFace under simulated Gaussian label noise with corruption levels controlled by std$_{max}$. Best-baseline entries are selected separately for each metric within the corresponding group.

Ablation Study

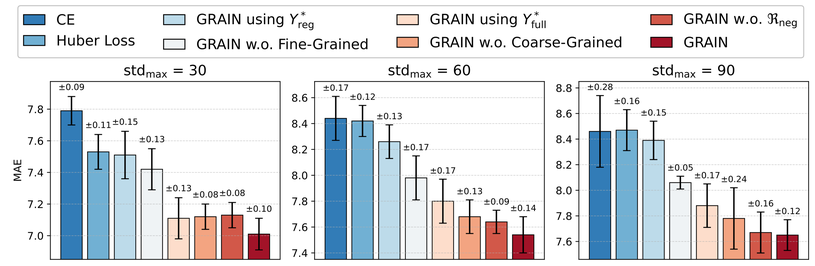


Figure 3: Ablation studies on UTKFace, highlighting the individual contributions of each related component.

Hyperparameter Sensitivity

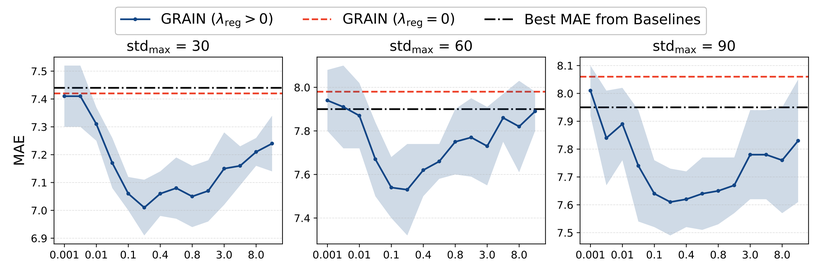


Figure 4: Hyperparameter sensitivity under different values of λ_{reg} on UTKFace, showing that GRAIN achieves strong performance within a reasonable range of λ_{reg}.

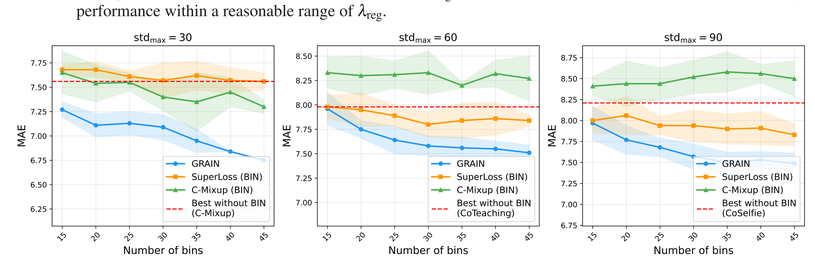


Figure B.5: Sensitivity analysis of the binning strategy on UTKFace, showing that the performance of GRAIN remains stable across different numbers of bins.

Contributions

1. Conduct **explicit residual learning** from a robust loss formulation, supported by statistical guarantees.
2. Couple **local correction and global ordering** through a histogram-based pseudo-label bridge.
3. Demonstrate **robustness across diverse noise settings**, with applications to both synthetic and real-world label noise data.